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py_neuromodulation: Signal processing and decoding for neural electrophysiological recordings

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Summary

Invasive brain signal decoding can revolutionize the clinical utility of neurotechnological therapies. Potential signal sources stem from electroencephalography (EEG), electrocorticography (ECoG) and local field potentials (LFP) recorded from deep brain stimulation (DBS) electrodes. The application of machine learning methods to these signals requires pre-processing and feature extraction – complex analyses, often lacking standardization and clear documentation. Here, we introduce py_neuromodulation, a toolbox designed for standardized signal processing, feature extraction, and decoding of electrophysiological data. All parameters are explicitly defined in dedicated settings and channel parameterization files. The framework processes both data streamed in real-time and from recordings on disk using the same pipeline. Additionally, a browser-based graphical user interface (GUI) enables intuitive usage and visualization of the processing pipeline without requiring any code modification.
By introducing py_neuromodulation, we aim to simplify and standardize the analysis of electrophysiological recordings, facilitating reproducibility and accessibility in the field of brain signal decoding. Our tool bridges the fields of neuroscience and neural engineering, providing machine learning and neurotechnology researchers with reproducible methods for the development of generalizable machine learning algorithms.

Introduction

Recently, deep learning foundation models were presented that showed a performance leap on many clinical relevant downstream tasks (Li et al., 2025; Yuan et al., 2024). Neural signal processing for machine learning and medical device engineering is a critical step for the advancement of the fields of neural decoding and brain computer interfaces. We, therefore, developed a neural signal processing, feature estimation and decoding toolbox py_neuromodulation, that includes several established and standardized pipelines that simplify processing of neural recordings.
Analysis of electrophysiological recordings is commonly started by multiple preprocessing steps: resampling, notch-filtering, normalization, re-referencing (subtraction of neighboring channels), filtering (lowpass, highpass, bandpass), or artifact rejection (Cohen, 2014). Next, the signals or features of interest are computed. Most commonly, signals are analyzed in the spectral domain. Here, different methods can be utilized, including Fast Fourier Transform, Welch transform, bandpass filtering, and others. In addition to spectral features, different feature

modalities such as waveform-shape (Cole & Voytek, 2017), periodic and aperiodic spectral parametrization (Donoghue et al., 2020), burst features (Tinkhauser et al., 2018), and others can be computed.

Statement of need

The variety of different processing parameters poses a challenge to select optimal methods for machine learning-based decoding, and second, hinders reproducibility, since often necessary parameters are not reported in scientific literature or reproducible from the derived data. *py_neuromodulation* utilizes two parametrization files: *settings.yaml* and *channels.tsv*, that include all required steps for reproducibility. It allows for a streamlined pipeline of pre-processing, feature estimation and decoding. The performance contribution of various pre-processing steps can be assessed in combination with different feature estimation modalities and machine learning model architectures.
Offline analysis often includes whole-recording normalization, non-causal filtering, or feature computation with test-set data leakage (Merk, Peterson, Köhler, et al., 2022), which limits real-time compatibility. In py_neuromodulation, the data-source is represented as a Python generator, and can be interchangeable for online and offline processing, meaning that there is no difference in handling data streamed directly from a brain implant, or stored from an archival dataset. This enables training of generalizable models that are directly applicable to online applications, using exactly the same algorithms. Finally, electrophysiological research entails multiple technical and conceptual hurdles that pose substantial entry barriers for scientists without software development training. By providing an intuitive GUI for parametrization and raw-data and feature visualization, we aim to lower the domain knowledge requirements for entering this field. Py_neuromodulation was already used in several scientific projects with various applications: ECoG and DBS-LFP movement decoding (Cavallo et al., 2025; Merk, Peterson, Lipski, et al., 2022), movement intention classification (Köhler et al., 2024), seizure classification (Merk et al., 2023), emotional valence decoding (Merk et al., 2023), gait symptom estimation in rodents (ELGarulli, 2025), amongst others.

Parametrization

A data-source can be specified to be either a two-dimensional array (channels x time) read through mne.io.read_raw supported recording standards or a LabStreamingLayer stream based on mne-lsl (Figure 1). If the data is passed as a simple numpy array, the sampling rate needs to be additionally specified. The parametrization **settings** and **channels** can be passed as additional optional parameters or automatically inferred. All parametrization is described in the GitHub documentation in detail Usage page. The main parameters are the *sampling_rate_features_hz* and *segment_length_ms*. The feature sampling rate defines in which interval features are computed, and the segment length defines the signal duration that is used for computation of each feature. For pre-processing we defined several methods that can be reordered freely: notch_filtering, raw_resampling, re_referencing (channel-specific common-average or individual bipolar re-referencing), raw_normalization and preprocessing_filter (lowpass, highpass, bandpass, bandstop). Additionally, a modifiable list of frequency ranges is passed, which are used for spectral feature computation. A list of currently implemented features is the following: raw_hjorth, return_raw, bandpass_filter, stft, fft, welch, sharpwave_analysis, Hjorth, line length, bispectrum, fooof, bursts, linelength, coherence, nolds, mne_connectivity and bispectrum.

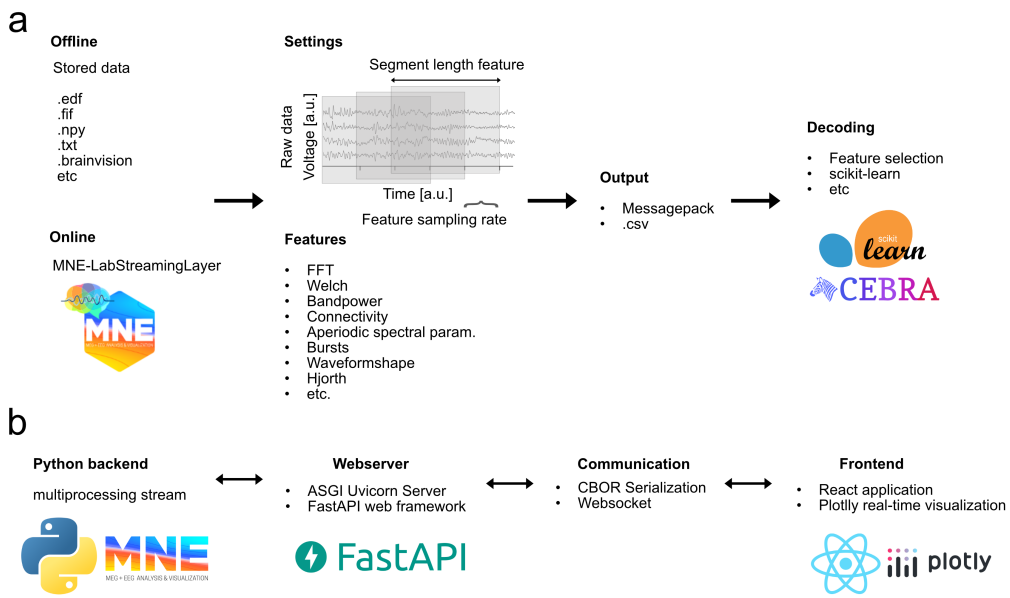

Figure 1: Schematic of py_neuromodulation .

Importantly, pre-processing and feature estimation can directly affect performances of decoding models. We added a **decode** module that can be parametrized with a scikit-learn compatible model, different cross-validation strategies, performance metrices, and dimensionality reduction methods, such that the computed features can be directly used to evaluate decoding performance within the same pipeline. A pre-trained model can also be passed to directly generate decoding outputs.

Graphical User Interface (GUI)

Finally, to simplify electrophysiological analysis and decoding, we added a React-based frontend application, which runs within an ASGI (Asynchronous Server Gateway Interface) uvicorn server and communicates through FastAPI. Through this frontend, the same settings as in the Python backend can be modified (Figure 2), a stream can then be selected from an offline file or via LabStreamingLayer, and raw data and features can be visualized in real time.

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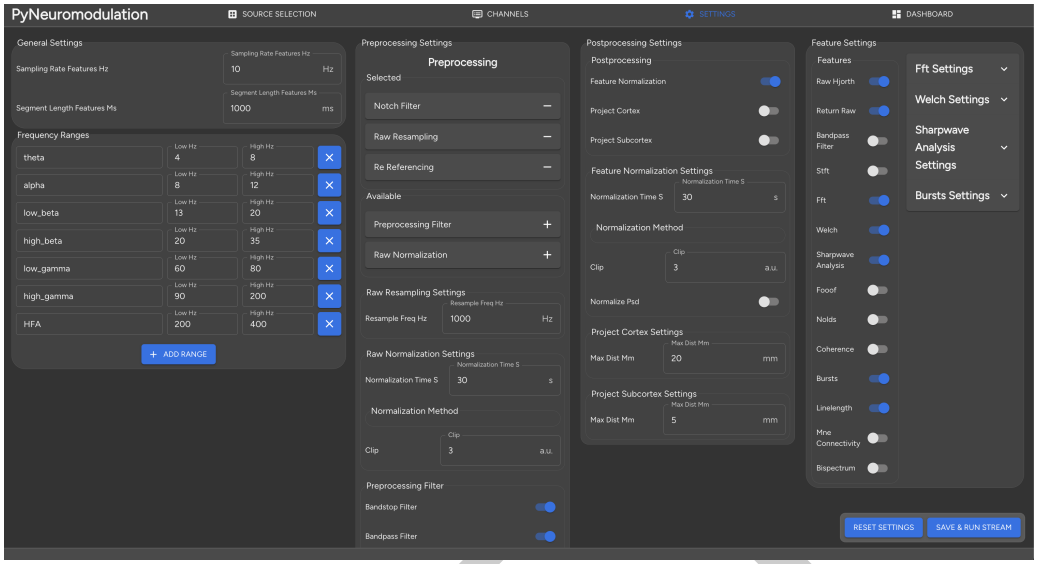

Figure 2: Frontend parametrization page representing `settings.yaml` configurations for pre-processing, feature estimation and post-processing.

Conclusion

In summary, py_neuromodulation provides a comprehensive, standardized framework for electrophysiological signal processing and neural decoding, addressing existing limitations in reproducibility and parameter documentation. Its unified pipeline supports both real-time and offline analyses, accompanied by an intuitive graphical interface to lower technical barriers in neural research. The successful use of py_neuromodulation across diverse applications highlights its potential as a broadly applicable tool for the analysis and machine-learning based decoding of electrophysiological data.

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